Answer **all** the questions in the spaces provided.

1 Ball A falls vertically in air from rest. The variation with time *t* of the distance *d* moved by the ball is shown in Fig. 1.1.

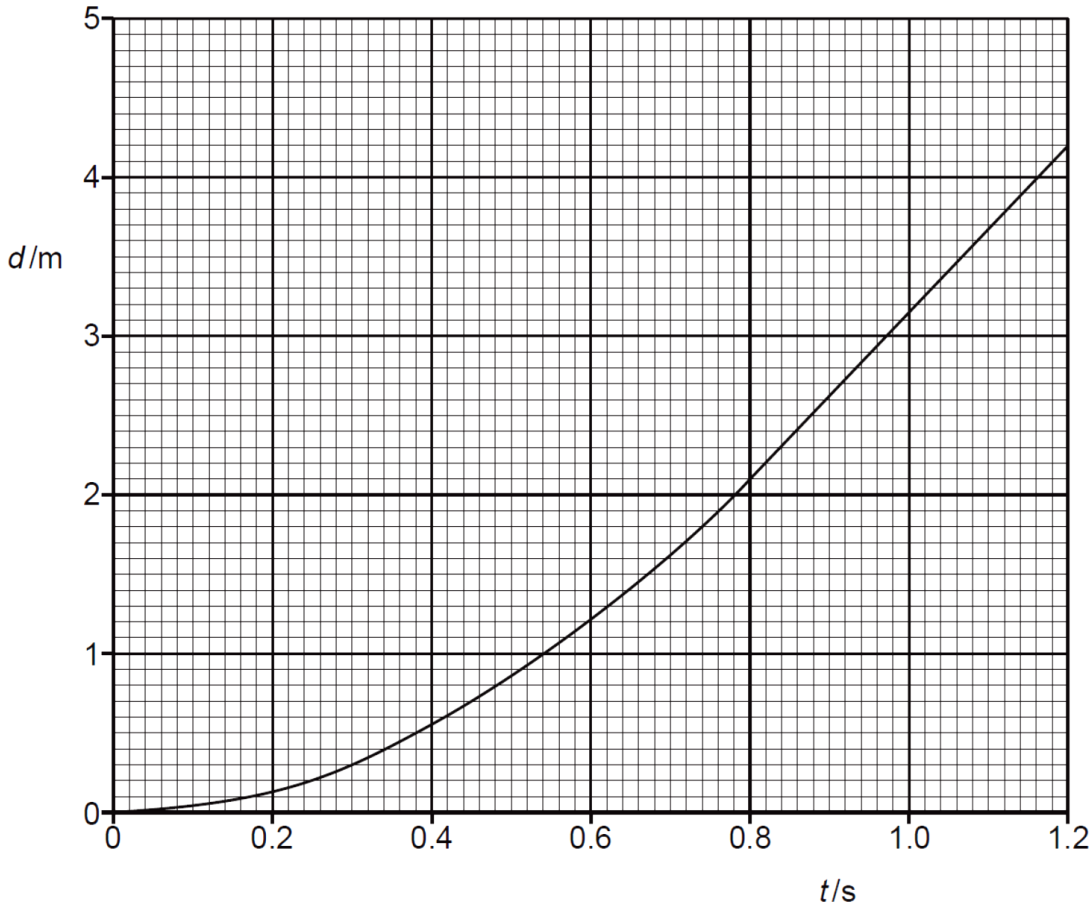

Fig. 1.1

(a) By reference to Fig. 1.1, explain how it can be deduced that air resistance is not negligible.

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……..

…….....[2]

(b) Use Fig. 1.1 to determine

(i) the speed of the ball at a time of 0.40 s after it has been released,

speed = ……. m s^{-1} [2]

(ii) the average speed of the ball in the first 0.40 s after it has been released.

average speed = …… m s^{-1} [1]

(c) Ball A is replaced by ball B which experiences negligible air resistance.

(i) Calculate the distance travelled by ball B after falling for 0.40 s from rest.

distance = ……m [1]

(ii) On Fig. 1.1, sketch a graph to show the variation with time *t* of the distance *d* moved by ball B after falling from rest. Label the graph P. [3]